CEE6501 — Written Assignment, Week 10

Assigned: 03/20/2026 (Week 10)

Due: 03/30/2026

Canvas Submission Link: https://gatech.instructure.com/courses/517856/assignments/2320544

Logistics

📄 Assignment Format

This is a **written assignment**, not a coding assignment.

- Complete the assignment **by hand or typed up**
- You may write:
  - in a markdown or jupyter file
  - on paper
  - on a tablet
  - using a stylus or digital note-taking app
- After completing the assignment, **scan or export your work** and upload it to Canvas

You may use **any format you prefer**, but your submission **must be neat and legible*

📤 Submission Instructions

- Upload **one file** to Canvas
- Accepted formats: **PDF, PNG, or JPG**
- Make sure all pages are clearly visible and in the correct order

✅ Checklist Before Submitting

- [ ] Assignment completed by hand
- [ ] Writing is neat and legible
- [ ] Title block provided on each page
- [ ] All questions answered
- [ ] File uploaded successfully to Canvas

🤝 Collaboration / AI tools

You may discuss concepts with classmates and you may use AI tools to help you learn, but **your submitted work must be written by you and you must understand it**. If you used outside help, add a short note in the final reflection cell.

Revisit Frame Structure in `A7_written.md`

Using your solution to Question 2 in `A7_written.md` as the starting point, now apply the methods from Week 9 for support displacements, temperature, and fabrication / fit-up forces.

You should treat this as an extension of your earlier DSM solution, not as a completely new problem. In particular, you may use your previously assembled stiffness matrix, DOF partitioning, and element information from your `A7_written.md` solution. For this assignment, the new effects enter through the **right-hand side of the equilibrium equations**, so you do not need to reassemble the global stiffness matrix.

Hand-written solutions are acceptable — but they must be **neat, clearly organized, and easy to follow**. Clearly label all steps, vectors, and intermediate results.

If possible, submit a clean solution in **Markdown** or a **Jupyter notebook**. A notebook allows you to structure your work clearly, embed images, and present your solution as a short technical report that walks through your reasoning step by step.

Submit / report

For each of the question below, submit the following:

- The modified right-hand-side vector(s) used in the solution
- Global displacements at the **free DOFs**
- Global reactions at the **restrained DOFs**
- A **global equilibrium check**
- Local element end forces
- Local element end displacements
- A clear, neat sketch of the **deformed shape**

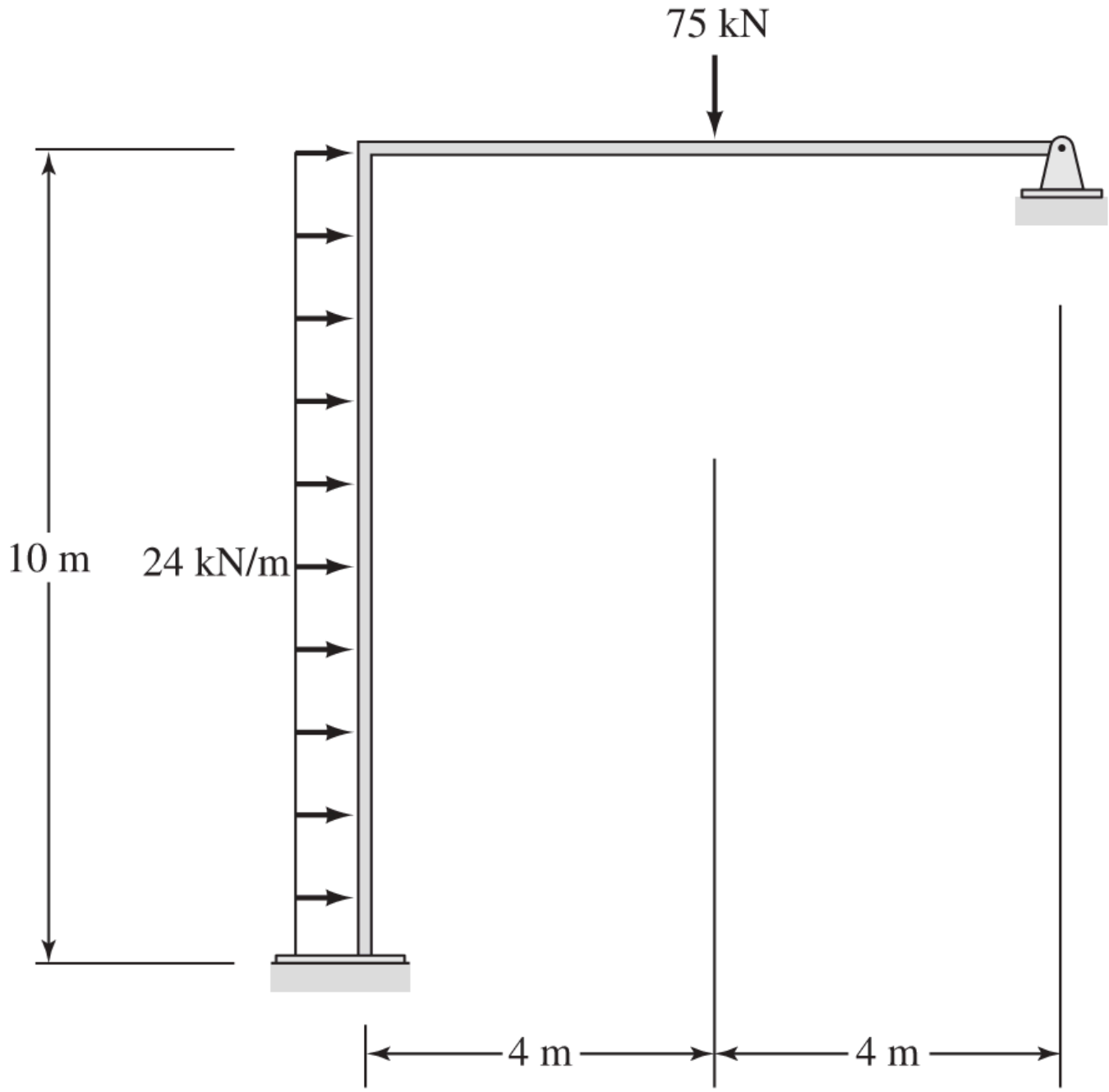
Numbering conventions

- Number global nodes **bottom-left → top-left → top-right**
- Use the Lecture 7 DOF convention (1-based):
  - $\mathrm{DOF}(j, u_x) = 3j - 2$
  - $\mathrm{DOF}(j, u_y) = 3j - 1$
  - $\mathrm{DOF}(j, \theta) = 3j$

Structural info

- E, A, I are constant
- $E = 200\ \mathrm{GPa}$
- $A = 4740\ \mathrm{mm}^2$
- $I = 22.2 \times 10^6\ \mathrm{mm}^4$

- use the α for steel as given in lecture:
  $\alpha_{\text{steel}} = 12 \times 10^{-6}\ /^{\circ}\text{C}$



Reuse the previous stiffness matrix and info

Integrated

$$\mathbf{u}_f = \begin{bmatrix} u_4 \\ u_5 \\ u_6 \\ u_9 \end{bmatrix}, \qquad \mathbf{u}_r = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_7 \\ u_8 \end{bmatrix}$$

$$\mathbf{F}_f = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{F}_r = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{K}_{ff} = \begin{bmatrix} 118553.28 & 0 & 266.4 & 0 \\ 0 & 94904.0625 & 416.25 & 416.25 \\ 266.4 & 416.25 & 3996 & 1110 \\ 0 & 416.25 & 1110 & 2220 \end{bmatrix}$$

$$\mathbf{K}_{fr} = \begin{bmatrix} -53.28 & 0 & 266.4 & -118500 & 0 \\ 0 & -94800 & 0 & 0 & -104.0625 \\ -266.4 & 0 & 888 & 0 & -416.25 \\ 0 & 0 & 0 & 0 & -416.25 \end{bmatrix}$$

$$\mathbf{K}_{rf} = \mathbf{K}_{fr}^T$$

$$\mathbf{K}_{rr} = \begin{bmatrix} 53.28 & 0 & -266.4 & 0 & 0 \\ 0 & 94800 & 0 & 0 & 0 \\ -266.4 & 0 & 1776 & 0 & 0 \\ 0 & 0 & 0 & 118500 & 0 \\ 0 & 0 & 0 & 0 & 104.0625 \end{bmatrix}$$

Element 1

$$\mathbf{k}'_{e1} = \begin{bmatrix} 94800 & 0 & 0 & -94800 & 0 & 0 \\ 0 & 53.28 & 266.4 & 0 & -53.28 & 266.4 \\ 0 & 266.4 & 1776 & 0 & -266.4 & 888 \\ -94800 & 0 & 0 & 94800 & 0 & 0 \\ 0 & -53.28 & -266.4 & 0 & 53.28 & -266.4 \\ 0 & 266.4 & 888 & 0 & -266.4 & 1776 \end{bmatrix}$$

$$\mathbf{T}_{e1} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{k}_{e1} = \mathbf{T}^T \, \mathbf{k}'_{e1} \, \mathbf{T} = \begin{bmatrix} 53.28 & 0 & -266.4 & -53.28 & 0 & -266.4 \\ 0 & 94800 & 0 & 0 & -94800 & 0 \\ -266.4 & 0 & 1776 & 266.4 & 0 & 888 \\ -53.28 & 0 & 266.4 & 53.28 & 0 & 266.4 \\ 0 & -94800 & 0 & 0 & 94800 & 0 \\ -266.4 & 0 & 888 & 266.4 & 0 & 1776 \end{bmatrix}$$

$$\mathbf{Q}_{e1}^{F,load} = \begin{bmatrix} 0 \\ 120 \\ 200 \\ 0 \\ 120 \\ -200 \end{bmatrix} \qquad \rightarrow \qquad \mathbf{F}_{e1}^{F,load} = \mathbf{T}_{e1}^T \mathbf{Q}_{e1}^{F,load} = \begin{bmatrix} -120 \\ 0 \\ 200 \\ -120 \\ 0 \\ -200 \end{bmatrix} \qquad (\text{kN},\ \text{kN},\ \text{kN} \cdot \text{m},\ \text{kN},\ \text{kN})$$

Element 2

$$\mathbf{k}_{e2} = \mathbf{k}'_{e2} = \begin{bmatrix} 118500 & 0 & 0 & -118500 & 0 & 0 \\ 0 & 104.0625 & 416.25 & 0 & -104.0625 & 416.25 \\ 0 & 416.25 & 2220 & 0 & -416.25 & 1110 \\ -118500 & 0 & 0 & 118500 & 0 & 0 \\ 0 & -104.0625 & -416.25 & 0 & 104.0625 & -416.25 \\ 0 & 416.25 & 1110 & 0 & -416.25 & 2220 \end{bmatrix}$$

$$\mathbf{Q}_{e2}^{F,load} = \begin{bmatrix} 0 \\ 37.5 \\ 75 \\ 0 \\ 37.5 \\ -75 \end{bmatrix} = \mathbf{F}_{e2}^{F,load} \qquad (\text{kN},\ \text{kN},\ \text{kN} \cdot \text{m},\ \text{kN},\ \text{kN})$$

Question 1 — Support Displacements

Starting from your solution to Question 2 in `A7_written.md`, redo the problem assuming the following prescribed support motions:

- The **top-right pin support** moves **10 mm downward**
- The **top-right pin support** also moves **5 mm to the right**
- The **bottom-left support** undergoes a **counterclockwise rotation of 0.01 rad**

Solution:

$$\mathbf{u}_r = \begin{bmatrix} 0 \\ 0 \\ 0.01 \\ 0.005 \\ -0.01 \end{bmatrix}, \qquad \mathbf{F}_f^F = \begin{bmatrix} -120 \\ 37.5 \\ -125 \\ -75 \end{bmatrix}$$

$$\mathbf{u}_f = \mathbf{K}_{ff}^{-1}(\mathbf{F}_f - \mathbf{F}_f^F - \mathbf{K}_{fr}\mathbf{u}_r) \qquad \rightarrow \qquad \begin{bmatrix} u_4 \\ u_5 \\ u_6 \\ u_9 \end{bmatrix} = K_{ff}^{-1} \begin{bmatrix} 709.836 \\ -38.540625 \\ 111.9575 \\ 70.8375 \end{bmatrix} = \begin{bmatrix} 0.00593846 \\ -0.00059439 \\ 0.02181937 \\ 0.02111055 \end{bmatrix} \ (\mathrm{m},\ \mathrm{rad})$$

Compute Global reactions at the restrained DOFs

$$\mathbf{F}_r^F = \begin{bmatrix} -120 \\ 0 \\ 200 \\ 0 \\ 37.5 \end{bmatrix}$$

$$\mathbf{F}_r = \mathbf{K}_{rf}\mathbf{u}_f + \mathbf{K}_{rr}\mathbf{u}_r + \mathbf{F}_r^F \qquad \rightarrow \qquad \begin{bmatrix} R_{x1} \\ R_{y1} \\ M_1 \\ R_{x3} \\ R_{y3} \end{bmatrix} = \begin{bmatrix} -128.79308 \\ 56.34835 \\ 238.71761 \\ -111.20692 \\ 18.65165 \end{bmatrix} \ (\mathrm{kN},\ \mathrm{kN},\ \mathrm{kN}\cdot\mathrm{m},\ \mathrm{kN},\ \mathrm{kN})$$

Global equilibrium check

Vertical force equilibrium ($+y$ upward)

External vertical load:

$$\sum F_y^{\text{ext}} = -75 \text{ kN}$$

Sum of vertical reactions:

$$\sum R_y = R_{y1} + R_{y3} = 56.34835 + 18.65165 = 75.00000 \text{ kN}$$

Therefore,

$$\sum R_y + \sum F_y^{\text{ext}} = 75.00000 - 75.00000 = 0$$

Horizontal force equilibrium ($+x$ to the right)

Column horizontal UDL resultant:

$$W = wL_{12} = 24(10) = 240 \text{ kN}$$

Sum of horizontal reactions:

$$\sum R_x = R_{x1} + R_{x3} = -128.79308 - 111.20692 = -240.00000 \text{ kN}$$

Therefore,

$$\sum R_x + \sum F_x^{\text{ext}} = -240.00000 + 240.00000 = 0$$

Moment equilibrium about Node 1 (CCW $+$)

External moments about Node 1:

- Column UDL resultant 240 kN at y = 5 m:

$$M_w = -240(5) = -1200 \text{ kN} \cdot \text{m}$$

- Beam point load 75 kN at x = 4 m:

$$M_P = -75(4) = -300 \text{ kN} \cdot \text{m}$$

Thus,

$$\sum M_{(1)}^{\text{ext}} = M_w + M_P = -1200 - 300 = -1500 \text{ kN} \cdot \text{m}$$

Reaction moments about Node 1:

- Fixed-end moment at Node 1:

$$M_1 = 238.71761 \text{ kN} \cdot \text{m}$$

- Reactions at Node 3 located at ((x,y)=(8,10)):

$$M_{(R_{x3}, R_{y3})} = xR_{y3} - yR_{x3} = 8(18.65165) - 10(-111.20692) = 1261.28239 \text{ kN} \cdot \text{m}$$

So,

$$\sum M_{(1)}^{\text{react}} = M_1 + (xR_{y3} - yR_{x3}) = 238.71761 + 1261.28239 = 1500.00000 \text{ kN} \cdot \text{m}$$

Therefore,

$$\sum M_{(1)}^{\text{react}} + \sum M_{(1)}^{\text{ext}} = 1500.00000 - 1500.00000 = 0$$

Local element end displacements

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0.01 \\ 0.00593846 \\ -0.00059439 \\ 0.02181937 \\ 0.005 \\ -0.01 \\ 0.02111055 \end{bmatrix}$$

Element 1

The element global displacement vector is

$$\mathbf{u}_{e1} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0.01 \\ 0.00593846 \\ -0.00059439 \\ 0.02181937 \end{bmatrix}$$

Using the transformation for the vertical member,

$$\mathbf{u}'_{e1} = \mathbf{T}_{e1}\mathbf{u}_{e1} = \begin{bmatrix} 0 \\ 0 \\ 0.01 \\ -0.00059439 \\ -0.00593846 \\ 0.02181937 \end{bmatrix}$$

Element 2

Since Element 2 is horizontal, local and global coordinates are the same:

$$\mathbf{u}'_{e2} = \mathbf{u}_{e2} = \begin{bmatrix} u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} = \begin{bmatrix} 0.00593846 \\ -0.00059439 \\ 0.02181937 \\ 0.005 \\ -0.01 \\ 0.02111055 \end{bmatrix}$$

Local element end forces

Use

$$\mathbf{Q}_e = \mathbf{k}'_e\mathbf{u}'_e + \mathbf{Q}^F_e$$

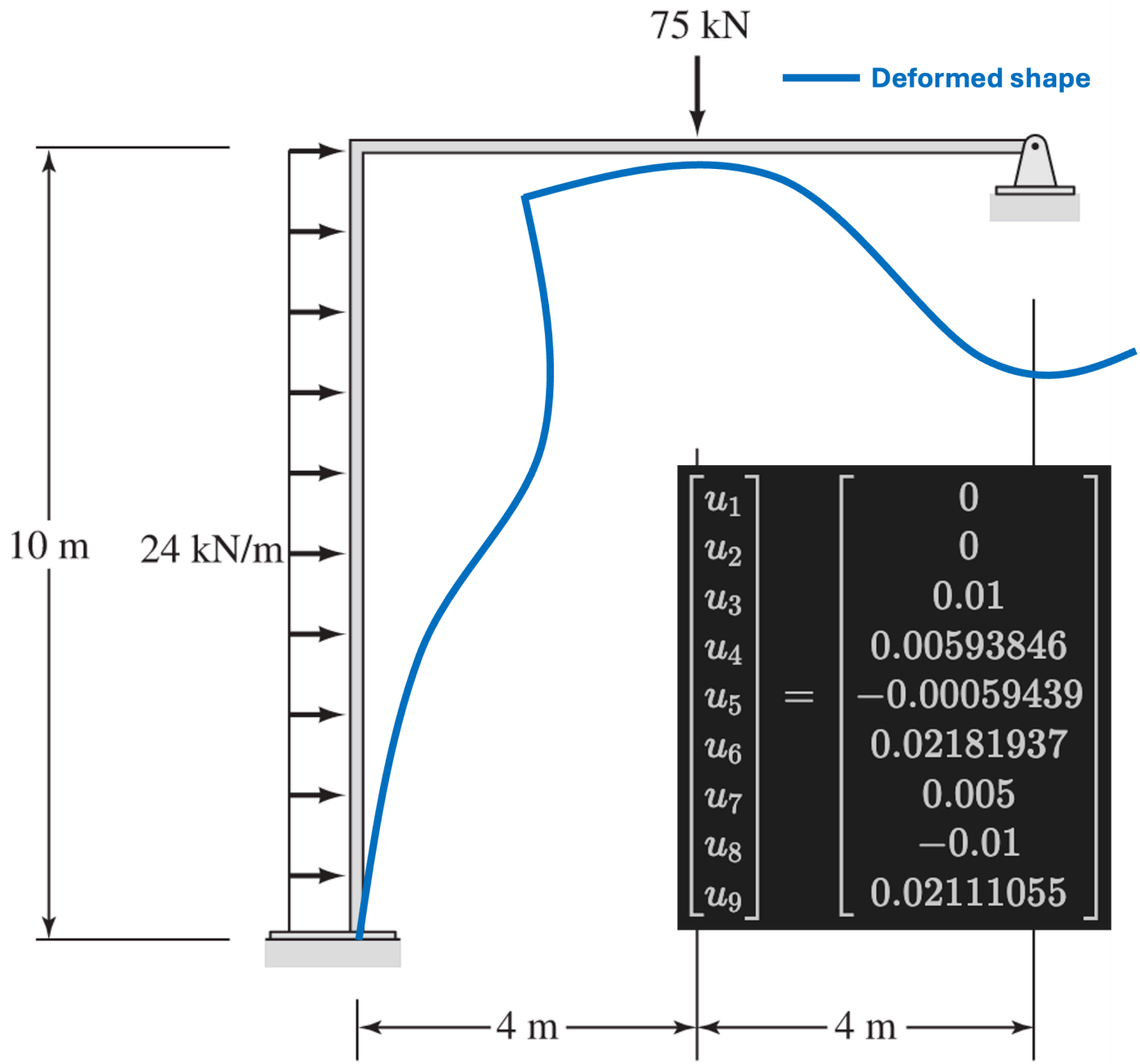
Element 1

$$\mathbf{Q}_{e1} = \mathbf{k}'_{e1}\mathbf{u}'_{e1} + \mathbf{Q}^F_{e1} = \begin{bmatrix} 56.34835 \\ 128.79308 \\ 238.71761 \\ -56.34835 \\ 111.20692 \\ -150.78679 \end{bmatrix} (\mathrm{kN},\ \mathrm{kN},\ \mathrm{kN}\cdot\mathrm{m})$$

Element 2

$$\mathbf{Q}_{e2} = \mathbf{k}'_{e2}\mathbf{u}'_{e2} + \mathbf{Q}^F_{e2} = \begin{bmatrix} 111.20692 \\ 56.34835 \\ 150.78679 \\ -111.20692 \\ 18.65165 \\ 0 \end{bmatrix} (\mathrm{kN},\ \mathrm{kN},\ \mathrm{kN}\cdot\mathrm{m})$$

Deformed shape



$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0.01 \\ 0.00593846 \\ -0.00059439 \\ 0.02181937 \\ 0.005 \\ -0.01 \\ 0.02111055 \end{bmatrix}$$

Question 2 — Temperature Effects

Starting from your solution to Question 2 in `A7_written.md`, redo the problem assuming the following temperature changes:

- **Vertical member:** uniform temperature increase of **+10°C**
- **Horizontal member:** temperature gradient of **+15°C at the top** and **+5°C at the bottom**

Solution

Temperature fixed-end forces

$$\mathbf{Q}_e^{F,temp} = \begin{bmatrix} Q_{xb}^{F,temp} \\ Q_{yb}^{F,temp} \\ Q_{mb}^{F,temp} \\ Q_{xe}^{F,temp} \\ Q_{ye}^{F,temp} \\ Q_{me}^{F,temp} \end{bmatrix} = E\alpha \begin{bmatrix} A(T_b + T_t)/2 \\ 0 \\ I(T_b - T_t)/d \\ -A(T_b + T_t)/2 \\ 0 \\ -I(T_b - T_t)/d \end{bmatrix}$$

where:

- T_b = temperature at the **bottom** face of the section,
- T_t = temperature at the **top** face of the section,
- $d = 300 \text{ mm} = 0.3 \text{ m}$,
- $\alpha = 12 \times 10^{-6} \ ^\circ\text{C}^{-1}$.

Element 1: uniform temperature increase of +10°C

$$T_b = T_t = 10^\circ\text{C}$$

$$EA\alpha(10) = 948000(12 \times 10^{-6})(10) = 113.76 \text{ kN}$$

Thus,

$$\mathbf{Q}_{e1}^{F,temp} = E\alpha \begin{bmatrix} A(10) \\ 0 \\ 0 \\ -A(10) \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 113.76 \\ 0 \\ 0 \\ -113.76 \\ 0 \\ 0 \end{bmatrix}$$

For the vertical member, transforming to global coordinates gives

$$\mathbf{F}_{e1}^{F,temp} = \mathbf{T}_{e1}^T \mathbf{Q}_{e1}^{F,temp} = \begin{bmatrix} 0 \\ 113.76 \\ 0 \\ 0 \\ -113.76 \\ 0 \end{bmatrix}$$

Element 2: temperature gradient of +15°C at the top and +5°C at the bottom

$$T_t = 15^\circ\text{C}, \qquad T_b = 5^\circ\text{C} \qquad \rightarrow \qquad \frac{T_b + T_t}{2} = 10, \qquad T_b - T_t = 5 - 15 = -10^\circ\text{C}$$

Thus,

$$\mathbf{Q}_{e2}^{F,temp} = E\alpha \begin{bmatrix} A(10) \\ 0 \\ I(-10)/d \\ -A(10) \\ 0 \\ -I(-10)/d \end{bmatrix} = \begin{bmatrix} 113.76 \\ 0 \\ -1.776 \\ -113.76 \\ 0 \\ 1.776 \end{bmatrix}$$

Since Element 2 is horizontal, local and global coordinates are the same:

$$\mathbf{F}_{e2}^{F,temp} = \begin{bmatrix} 113.76 \\ 0 \\ -1.776 \\ -113.76 \\ 0 \\ 1.776 \end{bmatrix}$$

Assemble the temperature fixed-end force vector

Assembling both element contributions into the global 9-DOF vector gives

$$\mathbf{F}^{F,temp} = \begin{bmatrix} 0 \\ 113.76 \\ 0 \\ 113.76 \\ -113.76 \\ -1.776 \\ -113.76 \\ 0 \\ 1.776 \end{bmatrix}$$

Reordered into free and restrained parts:

$$\mathbf{F}_f^{F,temp} = \begin{bmatrix} 113.76 \\ -113.76 \\ -1.776 \\ 1.776 \end{bmatrix}, \qquad \mathbf{F}_r^{F,temp} = \begin{bmatrix} 0 \\ 113.76 \\ 0 \\ -113.76 \\ 0 \end{bmatrix}$$

Solve global displacements at the free DOFs

The free-DOF equation is

$$\mathbf{u}_f = \mathbf{K}_{ff}^{-1}\left(\mathbf{F}_f - \mathbf{F}_f^{F,load} - \mathbf{F}_f^{F,temp}\right)$$

Since $\mathbf{F}_f = \mathbf{0}$,

$$\mathbf{u}_f = \mathbf{K}_{ff}^{-1}\left(-\begin{bmatrix}-120\\37.5\\-125\\-75\end{bmatrix} - \begin{bmatrix}113.76\\-113.76\\-1.776\\1.776\end{bmatrix}\right)$$

$$\rightarrow \qquad \begin{bmatrix}u_4\\u_5\\u_6\\u_9\end{bmatrix} = \mathbf{K}_{ff}^{-1}\begin{bmatrix}6.24\\76.26\\126.776\\73.224\end{bmatrix} = \begin{bmatrix}-6.1649\times10^{-6}\\6.0199\times10^{-4}\\2.61669\times10^{-2}\\1.97875\times10^{-2}\end{bmatrix} \qquad (\mathrm{m},\ \mathrm{rad})$$

Thus,

$$u_4 = -6.1649\times10^{-6}\ \mathrm{m}$$

$$u_5 = 6.0199\times10^{-4}\ \mathrm{m}$$

$$u_6 = 2.61669\times10^{-2}\ \mathrm{rad}$$

$$u_9 = 1.97875\times10^{-2}\ \mathrm{rad}$$

Compute global reactions at the restrained DOFs

The restrained-DOF reactions are

$$\mathbf{F}_r = \mathbf{K}_{rf}\mathbf{u}_f + \mathbf{F}_r^F + \mathbf{F}_r^{F,temp}$$

$$\rightarrow \qquad \begin{bmatrix}R_{x1}\\R_{y1}\\M_1\\R_{x3}\\R_{y3}\end{bmatrix} = \begin{bmatrix}-126.97054\\56.69115\\223.23458\\-113.02946\\18.30885\end{bmatrix} \qquad (\mathrm{kN},\ \mathrm{kN},\ \mathrm{kN\cdot m},\ \mathrm{kN},\ \mathrm{kN})$$

Therefore,

- **Node 1**

$$R_{x1} = -126.97054\ \mathrm{kN}, \qquad R_{y1} = 56.69115\ \mathrm{kN}, \qquad M_1 = 223.23458\ \mathrm{kN\cdot m}$$

- **Node 3**

$$R_{x3} = -113.02946\ \mathrm{kN}, \qquad R_{y3} = 18.30885\ \mathrm{kN}$$

Global equilibrium check

Vertical force equilibrium ($+y$ upward)

External vertical load:

$$\sum F_y^{\text{ext}} = -75 \text{ kN}$$

Sum of vertical reactions:

$$\sum R_y = R_{y1} + R_{y3} = 56.69115 + 18.30885 = 75.00000 \text{ kN}$$

Therefore,

$$\sum R_y + \sum F_y^{\text{ext}} = 75.00000 - 75.00000 = 0$$

Horizontal force equilibrium ($+x$ to the right)

Column horizontal UDL resultant:

$$W = wL_{12} = 24(10) = 240 \text{ kN}$$

Sum of horizontal reactions:

$$\sum R_x = R_{x1} + R_{x3} = -126.97054 - 113.02946 = -240.00000 \text{ kN}$$

Therefore,

$$\sum R_x + \sum F_x^{\text{ext}} = -240.00000 + 240.00000 = 0$$

Moment equilibrium about Node 1 (CCW $+$)

External moments about Node 1:

- Column UDL resultant 240 kN at y = 5 m:

$$M_w = -240(5) = -1200 \text{ kN} \cdot \text{m}$$

- Beam point load 75 kN at x = 4 m:

$$M_P = -75(4) = -300 \text{ kN} \cdot \text{m}$$

Thus,

$$\sum M_{(1)}^{\text{ext}} = M_w + M_P = -1200 - 300 = -1500 \text{ kN} \cdot \text{m}$$

Reaction moments about Node 1:

- Fixed-end moment at Node 1:

$$M_1 = 223.23458 \text{ kN} \cdot \text{m}$$

- Reactions at Node 3 located at ((x,y)=(8,10)):

$$M_{(R_{x3},R_{y3})} = xR_{y3} - yR_{x3} = 8(18.30885) - 10(-113.02946) = 1276.76542 \text{ kN} \cdot \text{m}$$

So,

$$\sum M_{(1)}^{\text{react}} = M_1 + (xR_{y3} - yR_{x3}) = 223.23458 + 1276.76542 = 1500.00000 \text{ kN} \cdot \text{m}$$

Therefore,

$$\sum M_{(1)}^{\text{react}} + \sum M_{(1)}^{\text{ext}} = 1500.00000 - 1500.00000 = 0$$

Local element end displacements

Using the global displacement vector

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -6.1649 \times 10^{-6} \\ 6.0199 \times 10^{-4} \\ 2.61669 \times 10^{-2} \\ 0 \\ 0 \\ 1.97875 \times 10^{-2} \end{bmatrix}$$

Element 1

The element global displacement vector is

$$\mathbf{u}_{e1} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -6.1649 \times 10^{-6} \\ 6.0199 \times 10^{-4} \\ 2.61669 \times 10^{-2} \end{bmatrix}$$

Using the transformation for the vertical member,

$$\mathbf{u}'_{e1} = \mathbf{T}_{e1}\mathbf{u}_{e1} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.00060199 \\ 6.1649 \times 10^{-6} \\ 0.02616691 \end{bmatrix}$$

Element 2

Since Element 2 is horizontal, local and global coordinates are the same:

$$\mathbf{u}'_{e2} = \mathbf{u}_{e2} = \begin{bmatrix} u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} = \begin{bmatrix} -6.1649 \times 10^{-6} \\ 0.00060199 \\ 0.02616691 \\ 0 \\ 0 \\ 0.01978745 \end{bmatrix}$$

Local element end forces

Use

$$\mathbf{Q}_e = \mathbf{k}'_e \mathbf{u}'_e + \mathbf{Q}_e^F + \mathbf{Q}_e^{F,temp}$$

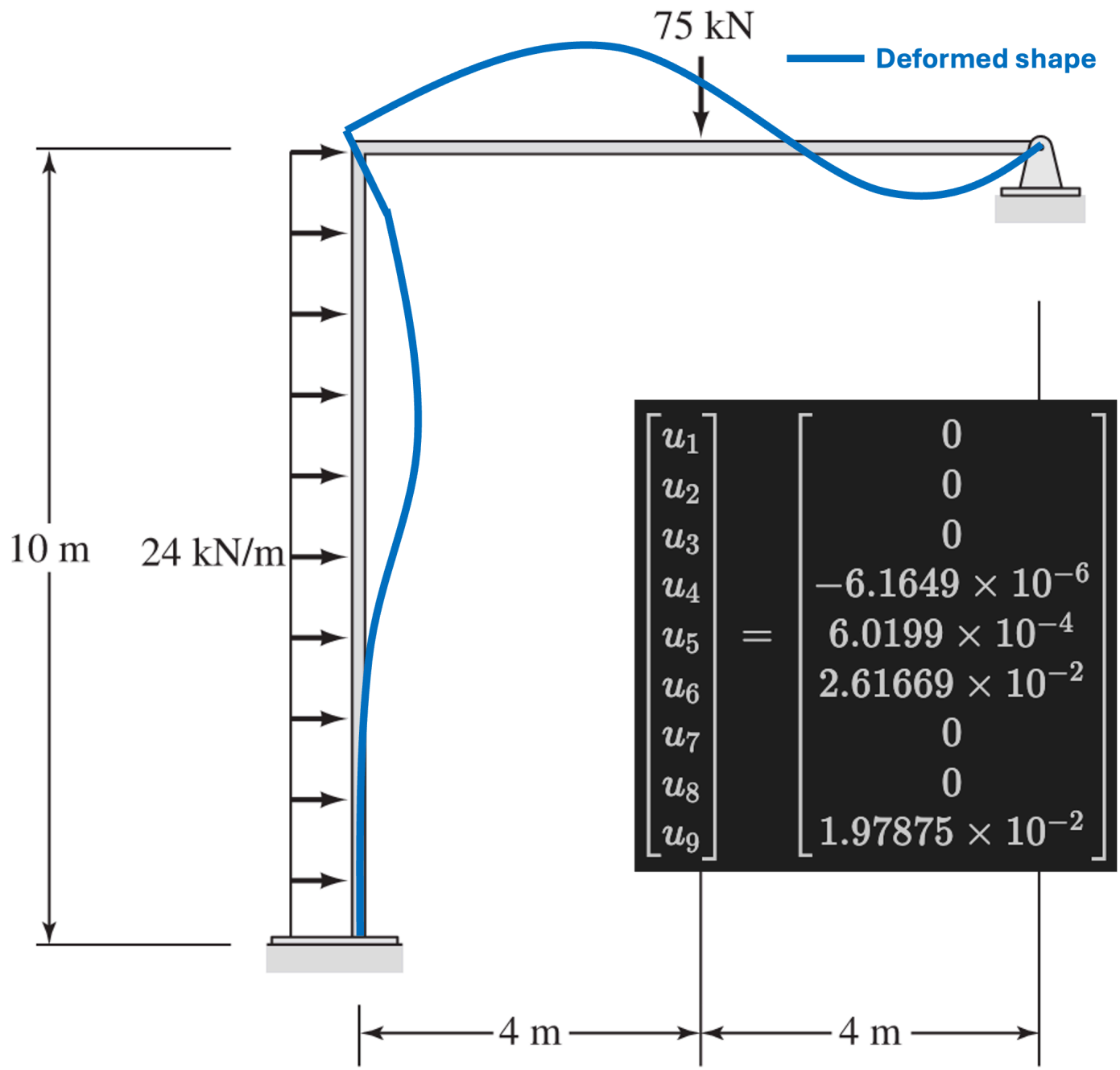
Element 1

$$\mathbf{Q}_{e1} = \mathbf{k}'_{e1} \mathbf{u}'_{e1} + \mathbf{Q}_{e1}^F + \mathbf{Q}_{e1}^{F,temp} = \begin{bmatrix} 56.69115 \\ 126.97054 \\ 223.23458 \\ -56.69115 \\ 113.02946 \\ -153.52920 \end{bmatrix} \qquad (\mathrm{kN},\ \mathrm{kN},\ \mathrm{kN} \cdot \mathrm{m})$$

Element 2

$$\mathbf{Q}_{e2} = \mathbf{k}'_{e2} \mathbf{u}'_{e2} + \mathbf{Q}_{e2}^F + \mathbf{Q}_{e2}^{F,temp} = \begin{bmatrix} 113.02946 \\ 56.69115 \\ 153.52920 \\ -113.02946 \\ 18.30885 \\ 0 \end{bmatrix} \qquad (\mathrm{kN},\ \mathrm{kN},\ \mathrm{kN} \cdot \mathrm{m})$$

Deformed shape



$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -6.1649 \times 10^{-6} \\ 6.0199 \times 10^{-4} \\ 2.61669 \times 10^{-2} \\ 0 \\ 0 \\ 1.97875 \times 10^{-2} \end{bmatrix}$$

Question 3 — Fabrication / Fit-up Errors

Starting from your solution to Question 2 in `A7_written.md`, redo the problem assuming the following fabrication errors:

- **Vertical member:** fabricated **3 mm too short**
- **Horizontal member:** fabricated **3 mm too short**

Solution

For a member fabricated too short, use the equivalent axial fixed-end-force vector in local coordinates:

$$\mathbf{Q}_e^{F,\text{error}} = \begin{bmatrix} Q_{xb}^{F,\text{error}} \\ Q_{yb}^{F,\text{error}} \\ Q_{mb}^{F,\text{error}} \\ Q_{xe}^{F,\text{error}} \\ Q_{ye}^{F,\text{error}} \\ Q_{me}^{F,\text{error}} \end{bmatrix} = EA\left(\frac{-0.003}{L}\right)\begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

where

$$EA = 948000 \text{ kN}$$

$$\mathbf{F}_f^{F,\text{load}} = \begin{bmatrix} -120 \\ 37.5 \\ -125 \\ -75 \end{bmatrix}, \qquad \mathbf{F}_r^{F,\text{load}} = \begin{bmatrix} -120 \\ 0 \\ 200 \\ 0 \\ 37.5 \end{bmatrix}$$

and there are no prescribed support displacements, so

$$\mathbf{u}_r = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{F}_f = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Fabrication error fixed-end forces

Element 1: vertical member fabricated 3 mm too short

For Element 1,

$$L_1 = 10 \text{ m}$$

so

$$EA\left(\frac{-0.003}{L_1}\right) = 948000\left(\frac{-0.003}{10}\right) = -284.4 \text{ kN}$$

Thus,

$$\mathbf{Q}_{e1}^{F,\text{error}} = EA\left(\frac{-0.003}{L_1}\right)\begin{bmatrix}1\\0\\0\\-1\\0\\0\end{bmatrix} = \begin{bmatrix}-284.4\\0\\0\\284.4\\0\\0\end{bmatrix}$$

For the vertical member, transforming to global coordinates gives

$$\mathbf{F}_{e1}^{F,\text{error}} = \mathbf{T}_{e1}^{T}\mathbf{Q}_{e1}^{F,\text{error}} = \begin{bmatrix}0\\-284.4\\0\\0\\284.4\\0\end{bmatrix}$$

Element 2: horizontal member fabricated 3 mm too short

For Element 2,

$$L_2 = 8 \text{ m}$$

so

$$EA\left(\frac{-0.003}{L_2}\right) = 948000\left(\frac{-0.003}{8}\right) = -355.5 \text{ kN}$$

Thus,

$$\mathbf{Q}_{e2}^{F,\text{error}} = EA\left(\frac{-0.003}{L_2}\right)\begin{bmatrix}1\\0\\0\\-1\\0\\0\end{bmatrix} = \begin{bmatrix}-355.5\\0\\0\\355.5\\0\\0\end{bmatrix}$$

Since Element 2 is horizontal, local and global coordinates are the same:

$$\mathbf{F}_{e2}^{F,\text{error}} = \begin{bmatrix}-355.5\\0\\0\\355.5\\0\\0\end{bmatrix}$$

Assemble the fabrication error fixed-end force vector

Assembling both element contributions into the global 9-DOF vector gives

$$\mathbf{F}^{F,\text{error}} = \begin{bmatrix} 0 \\ -284.4 \\ 0 \\ -355.5 \\ 284.4 \\ 0 \\ 355.5 \\ 0 \\ 0 \end{bmatrix}$$

Reordered into free and restrained parts:

$$\mathbf{F}_f^{F,\text{error}} = \begin{bmatrix} -355.5 \\ 284.4 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{F}_r^{F,\text{error}} = \begin{bmatrix} 0 \\ -284.4 \\ 0 \\ 355.5 \\ 0 \end{bmatrix}$$

Solve global displacements at the free DOFs

The free-DOF equation is

$$\mathbf{u}_f = \mathbf{K}_{ff}^{-1}\left(\mathbf{F}_f - \mathbf{F}_f^{F,\text{load}} - \mathbf{F}_f^{F,\text{error}}\right)$$

Since $\mathbf{F}_f = \mathbf{0}$,

$$\mathbf{u}_f = \mathbf{K}_{ff}^{-1}\left(-\begin{bmatrix} -120 \\ 37.5 \\ -125 \\ -75 \end{bmatrix} - \begin{bmatrix} -355.5 \\ 284.4 \\ 0 \\ 0 \end{bmatrix}\right)$$

$$\rightarrow \qquad \begin{bmatrix} u_4 \\ u_5 \\ u_6 \\ u_9 \end{bmatrix} = \mathbf{K}_{ff}^{-1}\begin{bmatrix} 475.5 \\ -321.9 \\ 125 \\ 75 \end{bmatrix} = \begin{bmatrix} 0.00395391 \\ -0.00359855 \\ 0.02534020 \\ 0.02178841 \end{bmatrix} \qquad (\text{m, rad})$$

Thus,

$$u_4 = 0.00395391 \text{ m}$$

$$u_5 = -0.00359855 \text{ m}$$

$$u_6 = 0.02534020 \text{ rad}$$

$$u_9 = 0.02178841 \text{ rad}$$

Compute global reactions at the restrained DOFs

The restrained-DOF reactions are

$$\mathbf{F}_r = \mathbf{K}_{rf}\mathbf{u}_f + \mathbf{F}_r^{F,\text{load}} + \mathbf{F}_r^{F,\text{error}}$$

$$\rightarrow \qquad \begin{bmatrix} R_{x1} \\ R_{y1} \\ M_1 \\ R_{x3} \\ R_{y3} \end{bmatrix} = \begin{bmatrix} -126.96129 \\ 56.74281 \\ 223.55542 \\ -113.03871 \\ 18.25719 \end{bmatrix} \qquad (\text{kN},\ \text{kN},\ \text{kN}\cdot\text{m},\ \text{kN},\ \text{kN})$$

Therefore,

- **Node 1**

$$R_{x1} = -126.96129\ \text{kN}, \qquad R_{y1} = 56.74281\ \text{kN}, \qquad M_1 = 223.55542\ \text{kN}\cdot\text{m}$$

- **Node 3**

$$R_{x3} = -113.03871\ \text{kN}, \qquad R_{y3} = 18.25719\ \text{kN}$$

Global equilibrium check

Vertical force equilibrium ($+y$ upward)

External vertical load:

$$\sum F_y^{\text{ext}} = -75\ \text{kN}$$

Sum of vertical reactions:

$$\sum R_y = R_{y1} + R_{y3} = 56.74281 + 18.25719 = 75.00000\ \text{kN}$$

Therefore,

$$\sum R_y + \sum F_y^{\text{ext}} = 75.00000 - 75.00000 = 0$$

Horizontal force equilibrium ($+x$ to the right)

Column horizontal UDL resultant:

$$W = wL_{12} = 24(10) = 240\ \text{kN}$$

Sum of horizontal reactions:

$$\sum R_x = R_{x1} + R_{x3} = -126.96129 - 113.03871 = -240.00000\ \text{kN}$$

Therefore,

$$\sum R_x + \sum F_x^{\text{ext}} = -240.00000 + 240.00000 = 0$$

Moment equilibrium about Node 1 (CCW $+$)

External moments about Node 1:

- Column UDL resultant 240 kN at y = 5 m:

$$M_w = -240(5) = -1200 \text{ kN} \cdot \text{m}$$

- Beam point load 75 kN at x = 4 m:

$$M_P = -75(4) = -300 \text{ kN} \cdot \text{m}$$

Thus,

$$\sum M_{(1)}^{\text{ext}} = M_w + M_P = -1200 - 300 = -1500 \text{ kN} \cdot \text{m}$$

Reaction moments about Node 1:

- Fixed-end moment at Node 1:

$$M_1 = 223.55542 \text{ kN} \cdot \text{m}$$

- Reactions at Node 3 located at ((x,y)=(8,10)):

$$M_{(R_{x3}, R_{y3})} = xR_{y3} - yR_{x3} = 8(18.25719) - 10(-113.03871) = 1276.44458 \text{ kN} \cdot \text{m}$$

So,

$$\sum M_{(1)}^{\text{react}} = M_1 + (xR_{y3} - yR_{x3}) = 223.55542 + 1276.44458 = 1500.00000 \text{ kN} \cdot \text{m}$$

Therefore,

$$\sum M_{(1)}^{\text{react}} + \sum M_{(1)}^{\text{ext}} = 1500.00000 - 1500.00000 = 0$$

Local element end displacements

Using the global displacement vector

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.00395391 \\ -0.00359855 \\ 0.02534020 \\ 0 \\ 0 \\ 0.02178841 \end{bmatrix}$$

Element 1

The element global displacement vector is

$$\mathbf{u}_{e1} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.00395391 \\ -0.00359855 \\ 0.02534020 \end{bmatrix}$$

Using the transformation for the vertical member,

$$\mathbf{u}'_{e1} = \mathbf{T}_{e1}\mathbf{u}_{e1} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -0.00359855 \\ -0.00395391 \\ 0.02534020 \end{bmatrix}$$

Element 2

Since Element 2 is horizontal, local and global coordinates are the same:

$$\mathbf{u}'_{e2} = \mathbf{u}_{e2} = \begin{bmatrix} u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} = \begin{bmatrix} 0.00395391 \\ -0.00359855 \\ 0.02534020 \\ 0 \\ 0 \\ 0.02178841 \end{bmatrix}$$

Local element end forces

Use

$$\mathbf{Q}_e = \mathbf{k}'_e\mathbf{u}'_e + \mathbf{Q}_e^{F,\text{load}} + \mathbf{Q}_e^{F,\text{error}}$$

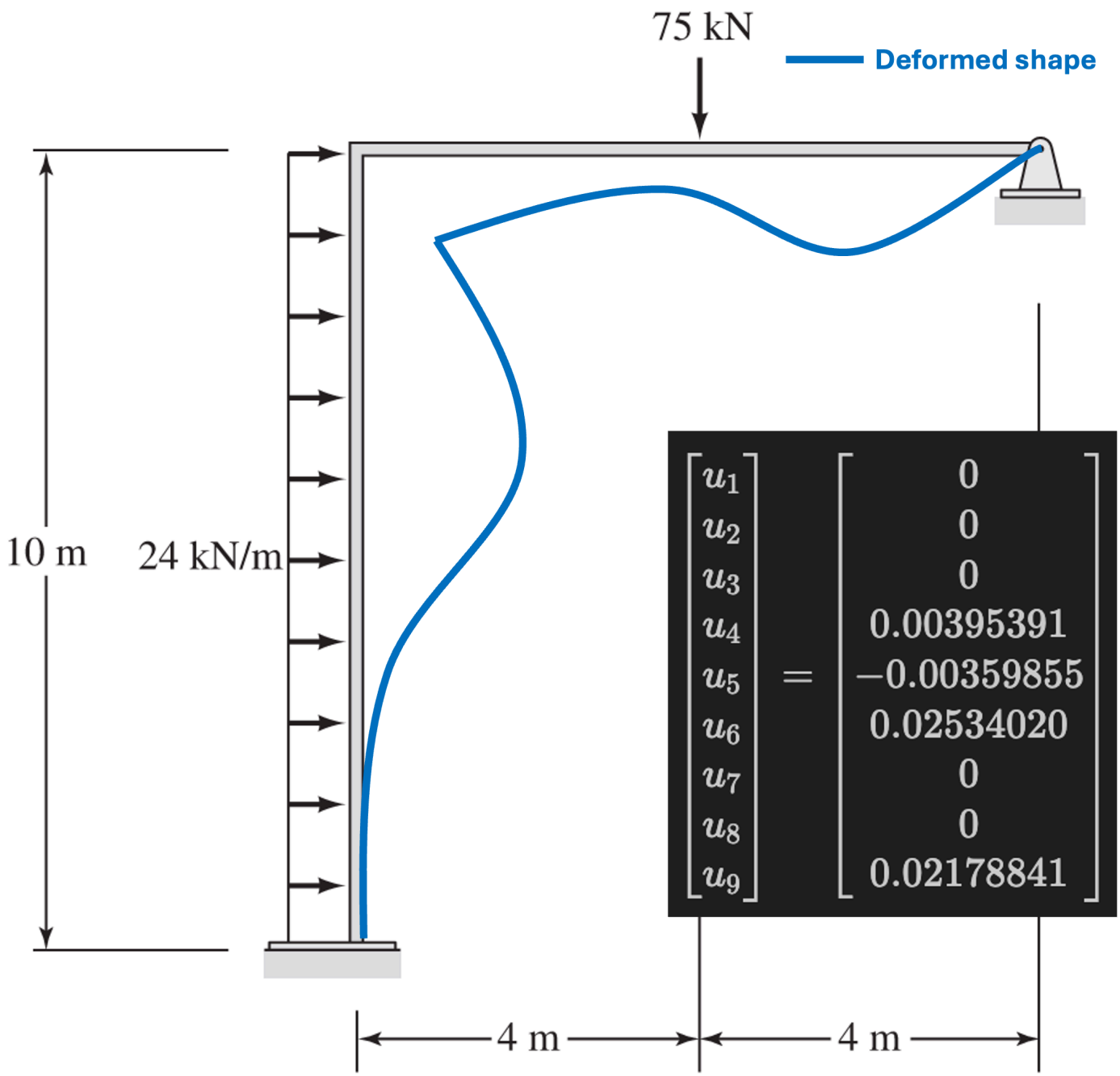
Element 1

$$\mathbf{Q}_{e1} = \mathbf{k}'_{e1}\mathbf{u}'_{e1} + \mathbf{Q}_{e1}^{F,\text{load}} + \mathbf{Q}_{e1}^{F,\text{error}} = \begin{bmatrix} 56.74281 \\ 126.96129 \\ 223.55542 \\ -56.74281 \\ 113.03871 \\ -153.94248 \end{bmatrix} \qquad (\text{kN},\ \text{kN},\ \text{kN}\cdot\text{m})$$

Element 2

$$\mathbf{Q}_{e2} = \mathbf{k}'_{e2}\mathbf{u}'_{e2} + \mathbf{Q}_{e2}^{F,\text{load}} + \mathbf{Q}_{e2}^{F,\text{error}} = \begin{bmatrix} 113.03871 \\ 56.74281 \\ 153.94248 \\ -113.03871 \\ 18.25719 \\ 0 \end{bmatrix} \qquad (\text{kN},\ \text{kN},\ \text{kN}\cdot\text{m})$$

Deformed shape



$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.00395391 \\ -0.00359855 \\ 0.02534020 \\ 0 \\ 0 \\ 0.02178841 \end{bmatrix}$$